Potentially new constructions

Low rank Grothendieck rings · Novelty shortlist and unresolved cases · 22 October 2026

To the best of our current knowledge, these seven constructions are promising candidates for new tensor categories. Their novelty up to tensor equivalence remains subject to independent verification.

Reported constructions to examine for novelty

m denotes maximum fusion multiplicity. Click a census ID for its construction source.

R6_4_249 Rank 6 · m = 4	**Sign-swap equivariantization** of the diagonal-fermion quotient of two identical half-E_6 categories. Known inputs: Aasen-Lake-Walker.
R6_4_322 Rank 6 · m = 4	**Mixed diagonal-fermion quotient** of Ad(E_6) and PSU(2)$_6$. Both input categories are already known.
R6_6_1712 Rank 6 · m = 6	**Relative Fibonacci center of TT$_3$.** The input TT$_3$ is the known dual even part of 2D2; see Edie-Michell, Definition 2.5.
R7_4_35 Rank 7 · m = 4	**Inversion descent and reflection Morita dual** of an orbifold of NG($C_4 \times C_4$,16). Input: Yu-Zhang.
R7_4_2665 Rank 7 · m = 4	**Analogous inversion/Morita construction** from NG(C_{16},16), with different invertibles and duality. Input: Budinski (2021).
R8_2_37 Rank 8 · m = 2	**Nonsplit C_2-extension of 2D2,** with an odd invertible a satisfying $a^2 = g$. Uses Grossman-Izumi-Snyder and Grossman.
R8_2_575 Rank 8 · m = 2	**Explicit Davydov-Runkel extension** of Kashina's self-dual 16-dimensional Hopf algebra. Simple dimensions: (1,1,1,1,2,2,2,4).

Known model added in the rank-seven extension

R7_5_5678 · Rank 7 · m = 5. Rep(AGL(1,7)) = Rep($C_7 \rtimes C_6$), of global dimension 42. The full fusion tensor is verified. This is a known representation category, not an additional novelty candidate.

OPEN: 7 fusion rings with no established categorification

R6_4_155 · Rank 6 · multiplicity 4. Generalized near-group ring R′(4): four invertibles forming $C_2 \times C_2$, and two mutually dual noninvertibles of dimension $4 + 3\sqrt{2}$. The ring already appears in Meng-Yu, Theorem 1.1.

R6_8_1660 · Rank 6 · multiplicity 8. Self-dual ring containing an Ising fusion subring. A further simple X has dimension $3 + 2\sqrt{2} + 2\sqrt{3} + \sqrt{6}$ and satisfies $X^2 = 1 + [3(1 + g) + 4\sigma]X$, where $\sigma^2 = 1 + g$.

Rank 7 · multiplicity 5: 0 unresolved rings. Of all 5,805 new rings, 5,804 are excluded and one is realized. Every supplied ring in this band has a recorded decision.

These 2 earlier remaining cases are open for general complex categorification. The rank-eight extension appears on the next page. Exact fusion data.

AI-generated research · independent verification pending.
Independent human verification and Lean formalization remain pending. Please treat the claims with caution.

Rank eight · multiplicity three

Accepted constructions and individually investigated candidates · 22 October 2026

3,527 new fusion rings. 3,506 general exclusions, 16 unitary constructions and 5 unresolved tensors. The earlier 973 records and their identifiers are preserved.

Explicit constructions

R8_3_274 · Phase-rotation C2 equivariantization of NG(C3,6)
R8_3_331 · NG(C3,3) x Vec(C2)
R8_3_334 · Reflection C2 equivariantization of NG(C3,6)
R8_3_342 · Rep(C13 semidirect C6)
R8_3_448 · Unitary Izumi–Xu Z2 extension (2221)
R8_3_458 · Bimodules over the conformal G2_3 to E6_1 algebra
R8_3_471 · NG(C3,3) x Fibonacci
R8_3_475 · Cyclic permutation equivariantization of Fibonacci cubed
R8_3_715 · Morita dual of the nonsplit unitary 2D2 extension
R8_3_724 · Rep(PSL(2,11))
R8_3_1456 · Ad(E8) x Vec(C2)
R8_3_3278 · E8 fusion category
R8_3_3314 · Ad(E8) x Fibonacci
R8_3_3342 · Twisted group-theoretical C(A6,omega,A5,psi) from a stable relative mu2 class
R8_3_721 · Bimodules over 1+rho in the C2 de-equivariantization of GH(C8)
R8_3_982 · Triality-zero component of E21 from the conformal embedding SU(3)_21 into E7_1

Each identification compares all 512 ordered fusion coefficients. Categorical existence uses the cited construction. The new applications do not by themselves prove that the resulting category is new up to tensor equivalence.

Remaining categorification candidates

R8_3_928 · R8_3_941 · R8_3_1154 · R8_3_1768 · R8_3_3472

The per-case audit distinguishes exact necessary-condition survivors, unsuccessful construction searches, ineligible tests and time limits. Remaining rings are extensively investigated exotic candidates in the documented screening sense; existence and novelty of a future realization remain unproved. Individual attempt records · Exact unresolved tensors.

AI-generated research · independent verification pending.
Exact replay checks stated finite certificates; new mathematical arguments still require human review.